

AI-Powered Enhanced EHR Imaging & Documentation System

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Objective

- Add a little bit of To automate and enhance Electronic Health Records (EHR) by integrating AI-powered imaging and documentation.
- To improve diagnostic accuracy through medical image enhancement using super-resolution techniques.
- To reduce manual effort in clinical documentation by generating structured notes automatically.
- To streamline ICD-10 mapping for faster, more reliable coding and billing.
- To deliver standardized, EMR-ready outputs (JSON/CSV) for seamless hospital integration.
- To deploy scalable solutions via cloud (Azure/Kubernetes) with a user-friendly interface for doctors.



Problem Statement

- Manual documentation & ICD-10 mapping → time-consuming, error-prone
- Poor image quality → low diagnostic accuracy
- Lack of standardization in EHR data
- Reduced clinical efficiency



Proposed Solution



- AI-powered automation for clinical notes & ICD-10 mapping
- Super-resolution imaging for diagnostic clarity
- Structured outputs (JSON/CSV) for EMR integration
- Cloud deployment for scalability

Milestone Journey

**Milestone 1: Data
Collection &
Preprocessing**

**Milestone 2:
Medical Imaging
Enhancement
(Real-ESRGAN,
PyTorch)**

**Milestone 3:
Clinical Note
Generation +
ICD-10 Mapping
(Azure OpenAI
GPT-4o-mini)**

**Milestone 4: API
+ UI Integration,
Dockerized
Deployment,
Cloud readiness**

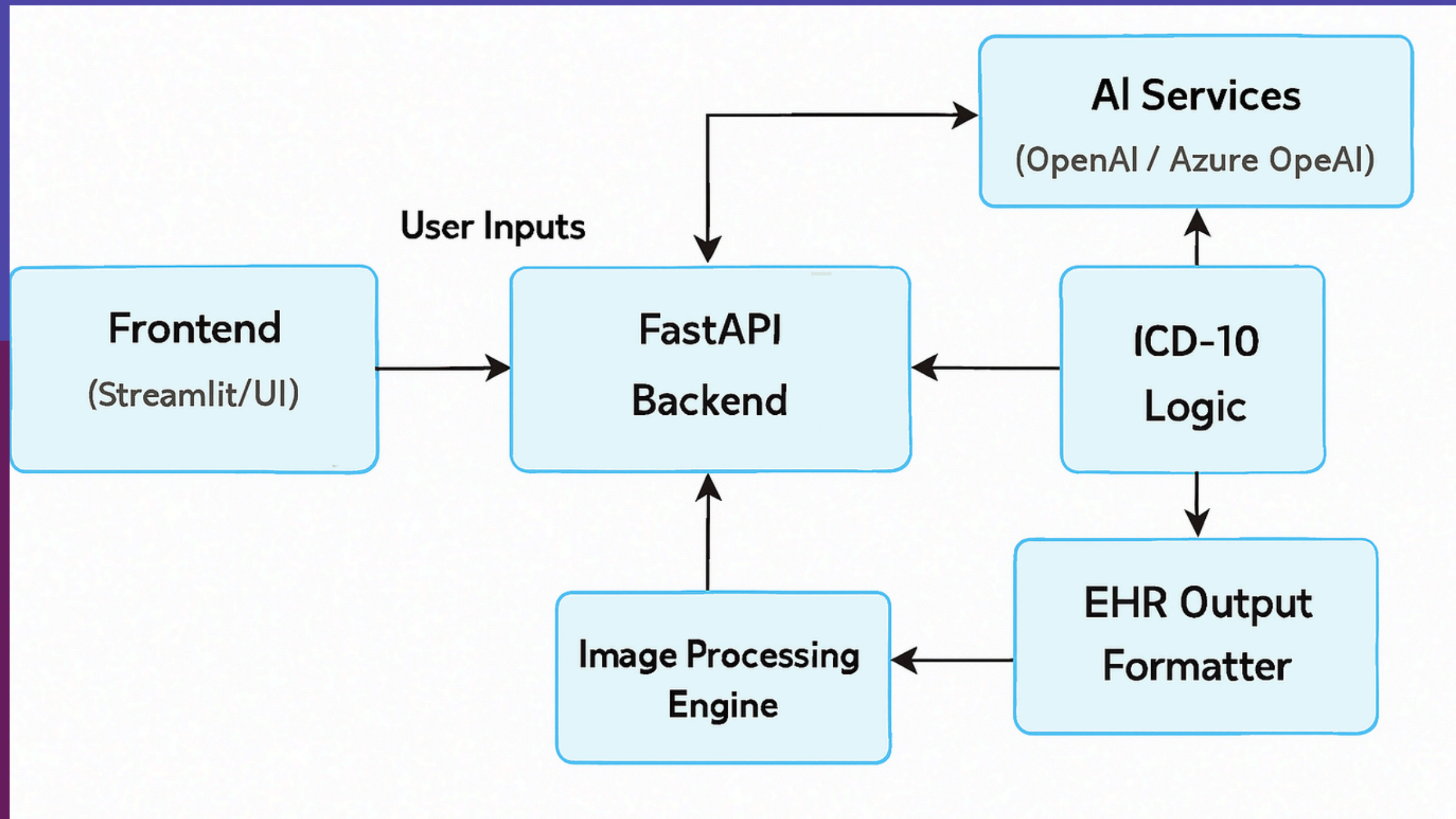




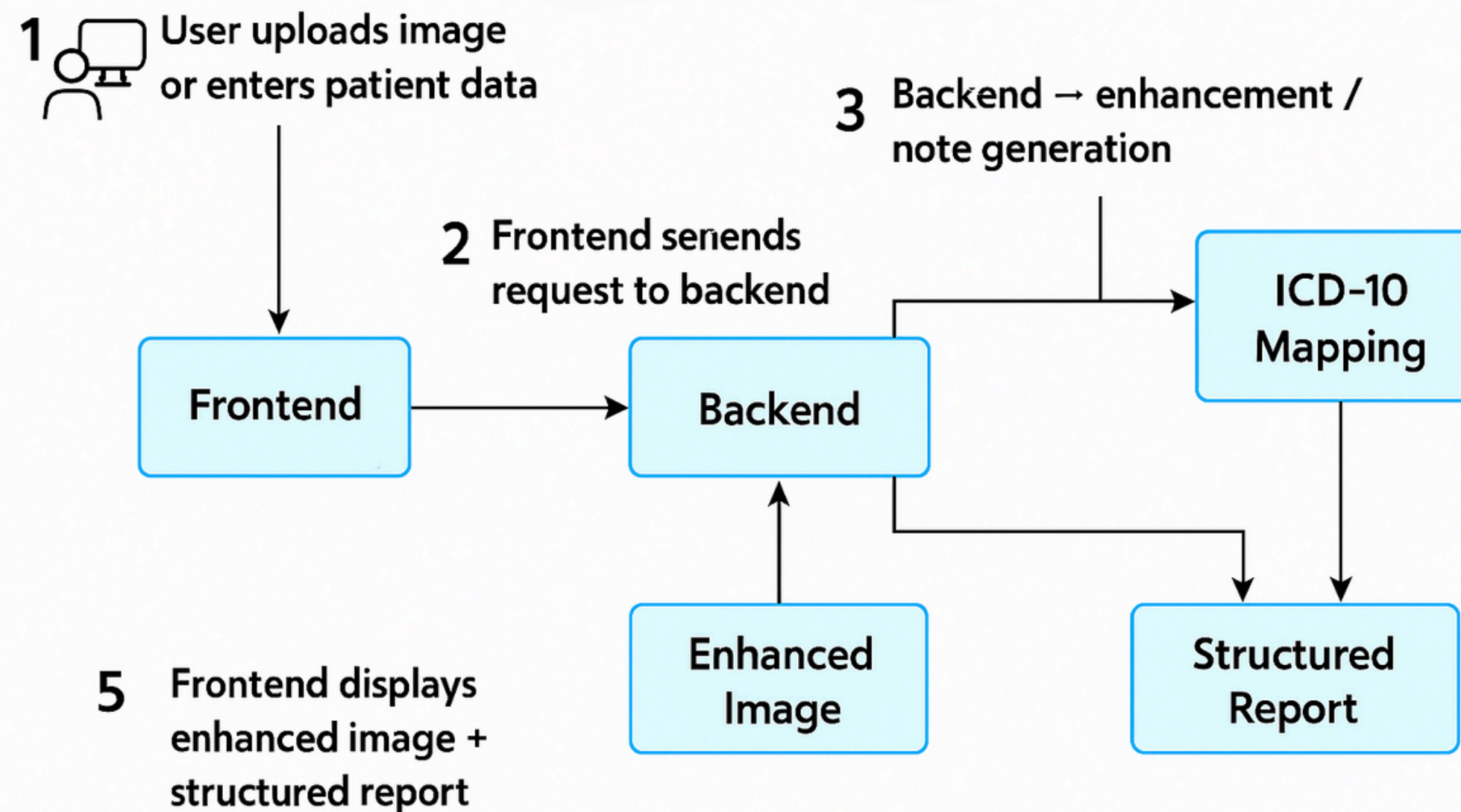
Tools & Technologies

- Python, FastAPI, Streamlit
- Azure OpenAI (GPT-4o-mini)
- PyTorch, Real-ESRGAN, OpenCV, PIL
- Pandas, JSON, CSV
- Docker, Kubernetes, Azure Cloud

Implementation Progress



Data Flow Diagram



Challenges Faced

1. Data Collection & Preprocessing (Milestone 1)

- Dataset inconsistencies: Missing categories (e.g., “No Tumor”) caused imbalance.
- Filename mismatches: Inconsistent MRI image naming complicated mapping.

2. Medical Imaging Enhancement (Milestone 2)

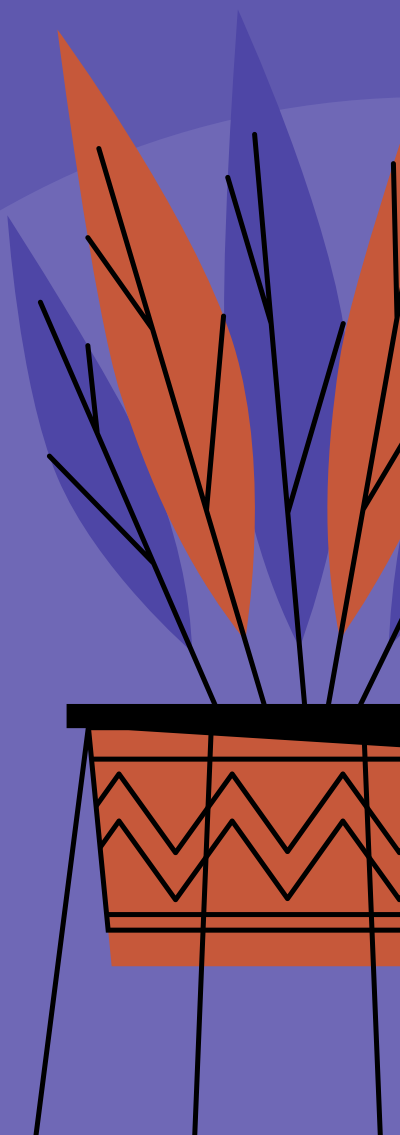
- Library conflicts: basicsr incompatibility with NumPy 2.0 → frequent crashes.
- Pretrained weight errors: Missing params_ema key in ESRGAN .pth files.
- Noise & resolution issues: Low-quality medical images required advanced super-resolution techniques.

3. Clinical Notes & ICD-10 Mapping (Milestone 3)

- Medical terminology complexity: AI struggled with nuanced clinical language.
- ICD-10 mismapping: Similar symptoms led to incorrect code assignments.
- Prompt engineering challenges: Needed refined prompts for accurate note generation.

4. Integration & Deployment (Milestone 4)

- API stability: Ensuring FastAPI endpoints handled large files and JSON inputs reliably.
- UI usability: Designing a doctor-friendly Streamlit interface with clarity and simplicity.
- Cloud deployment hurdles: Containerization, scaling, and Kubernetes orchestration required extra optimization.



Learnings & Solutions



1. Data Collection & Preprocessing (Milestone 1)

- Learning: Real-world datasets often contain inconsistencies and missing categories.
- Solution: Added missing “No Tumor” class and automated filename validation scripts to ensure dataset balance and reliability.

2. Medical Imaging Enhancement (Milestone 2)

- Learning: AI-based super-resolution models (Real-ESRGAN) significantly improve diagnostic clarity.
- Solution: Rebuilt RRDBNet in pure PyTorch to resolve library conflicts, applied denoising and resolution enhancement for clearer medical images.

3. Clinical Notes & ICD-10 Mapping (Milestone 3)

- Learning: Prompt engineering is critical for accurate medical text generation.
- Solution: Designed structured prompts and added post-processing validation to refine ICD-10 mapping accuracy and reduce mismatches.

4. Integration & Deployment (Milestone 4)

- Learning: Usability and scalability are as important as technical accuracy.
- Solution: Built a doctor-friendly Streamlit UI, containerized APIs with Docker, and prepared cloud deployment (Azure/Kubernetes) for real-world hospital integration.

Results & Achievements

- Enhanced medical images → clearer diagnostics
- Automated notes → standardized documentation
- ICD-10 mapping → accurate billing & coding
- Reduced documentation time → improved efficiency
- Export-ready structured outputs → EHR compatibility

